

TEK4090 – Introduction to Modern Control

Assignment 1

Due: September 18 2025, 13:15

1. (10 points) Suppose that $v_1, \dots, v_m$ is linearly independent in V , and that $w \in V$. Show that

$$v_1, \dots, v_m, w \text{ is linearly independent} \iff w \notin \text{span}(v_1, \dots, v_m). \quad (1)$$

Ans: I will show that $i) \rightarrow ii)$ and then that $ii) \rightarrow i)$

Assume $v_1, \dots, v_m, w$ is lin. indep.

Pf by contradiction:

$$w \in \text{span}(v_1, \dots, v_m)$$

$$w = \alpha_1 v_1 + \dots + \alpha_m v_m$$

$$0 = \alpha_1 v_1 + \dots + \alpha_m v_m + (-1)w$$

This contradicts that $v_1, \dots, v_m, w$ is linearly independent, because for any combination of indep. vectors, they must equal if and only if all coefficients $\alpha_1, \dots, \alpha_m = 0$. Since we have that the coefficient of w is equal to $-1 \neq 0$, we have that vectors $v_1, \dots, v_m, w$ are indep. and thus, $w \notin \text{span}(v_1, \dots, v_m)$

will now prove that $ii) \rightarrow i)$. Will also use proof by contradiction. Assume that $w \in \text{span}(v_1, \dots, v_m)$. This means that w can be written as a linear combination of the vectors in $\text{span}(v_1, \dots, v_m)$

$$w = \alpha_1 v_1 + \dots + \alpha_m v_m$$

This means that by definition, the list $(v_1, \dots, v_m, w)$ is dependent, and this is our contradiction. Thus, for $v_1, \dots, v_m, w$ to be independent, $w \notin \text{span}(v_1, \dots, v_m)$ has to be true

2. (10 points) Let $A, B \in \mathbb{R}^{n \times n}$, and define the *trace* of A as $\text{tr}(A) = \sum_{i=1}^n A_{ii}$. Show that

$$\frac{d\text{tr}(AB)}{dA} = B^T. \quad (2)$$

Hint: Find $\frac{d\text{tr}(AB)}{dA_{ij}}$.

Ans:

$$\begin{aligned} \text{tr}(AB) &= \sum_n \sum_{i=1}^n (AB)_{i,i} \\ &= \sum_n \sum_{i=1}^n A_{i,j} B_{j,i} \end{aligned}$$

Thus we get that

$$\begin{aligned} \frac{d \text{tr}(AB)}{dA_{i,j}} &= \frac{d(\sum_n \sum_{i=1}^n A_{i,j} B_{j,i})}{dA_{i,j}} \\ &= B_{j,i} \end{aligned}$$

And if we now replace $A_{i,j}$ with A in the denominator, we get

$$\frac{d(\sum_n \sum_{i=1}^n A_{i,j} B_{j,i})}{dA} = B^T$$

3. (15 points) Suppose that $T : V \mapsto V$ is a linear map, and suppose that $S : V \mapsto V$ is an invertible linear map.

- (a) Prove that T and $S^{-1}TS$ have the same eigenvalues
- (b) Describe the relationship between the eigenvectors of T and the eigenvectors of $S^{-1}TS$.

Ans: We want to prove that

$$T(v) = \lambda v = S^{-1}TS(v)$$

Matrix multiply both sides by S and get

$$ST(v) = S(\lambda v) = SS^{-1}TS(v)$$

$$SS^{-1}TS(v) = ITS(v)$$

$$T(S(v)) = \lambda S(v)$$

where I is the identity matrix

Since S is a linear invertible map, we get that $S(\lambda v) = \lambda S(v)$

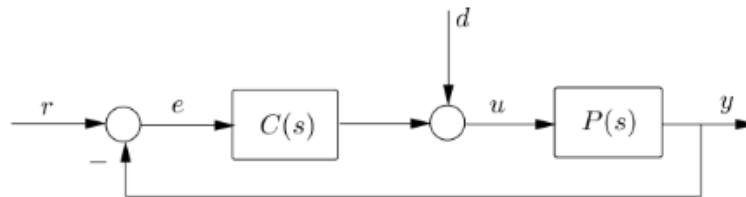
Now we set $v = S^{-1}(w)$ and $T(w) = \lambda w$

Then we have that $S(v) = w$

$$\begin{aligned} S^{-1}TS(v) &= S^{-1}T(w) \\ S^{-1}(\lambda w) &= \lambda S^{-1}(w) \\ &= \lambda v \end{aligned}$$

which is what we wanted to prove; namely that T and $S^{-1}TV$ have similar eigenvalues. This is correct assuming the maps T is linear and that S is linear and invertible

4. (30 points) Consider the feedback block diagram:



(a) Find the transfer function from d to y .

Ans:

The transfer function $T(s) = \frac{Y(s)}{D(s)}$ where

- $Y(s) = P(s)(C(s) + D(s))$
- $C(s) = U(s)E(s)$
- $E(s) = R(s) - Y(s)$

We are not interested in the error, so we set the reference to 0.

$$\begin{aligned} Y(s) &= P(s)U(s) + P(s)D(s) \\ &= P(s)[-Y(s)C(s)] + P(s)D(s) \\ Y(s)[1 + P(s)C(s)] &= P(s)D(s) \\ \frac{Y(s)}{D(s)} &= \frac{P(s)}{1 + P(s)C(s)} \end{aligned}$$

(b) Suppose the plant has transfer function

$$P(s) = \frac{(s+1)}{s^2 - 4s + 13} \quad (3)$$

What are the poles of $P(s)$? What can you infer about the stability of the (open-loop) plant $P(s)$?

Ans: We have an unstable open-loop plant, if at least one root of the denominator is located in the RHP. For $P(s) = \frac{N}{D}$, we have

$$s^2 - 4s + 13 = 0$$

In the abc formula, this gives

$$2 \pm j3$$

Since the real part of both roots are > 0 , we have instability of the plant $P(s)$.

(c) Suppose $C(s)$ is given by

$$C(s) = \frac{1}{s} \quad (4)$$

Does $C(s)$ stabilize the feedback system?

Ans: To find this out if controller $C(s)$ stabilizes the system, I find the roots of the characteristic equation

$$1 + C(s)P(s) = 0$$

If

- $C(s) = \frac{C_n}{C_d}$
- $P(s) = \frac{P_n}{P_d}$

I can formulate algebraically the new equation

$$C_d * P_d + C_n * P_n = 0$$

In matlab, this gives the following roots

- $2.0350 + 3.1849i$
- $2.0350 - 3.1849i$
- $-0.0700 + 0.0000i$

This confirms that $C(s)$ does not indeed stabilize the feedback system

Here is the script I used:

```
nC = [1];
dC = [1,0];

nP= [1,1];
dP = [1,-4,13];

numProd = conv(nD, nC);
denProd = conv(dC, dD);

ml = max(length(denProd), length(numProd));
denProd_padded = [zeros(1, ml - length(denProd)), denProd];
numProd_padded = [zeros(1, ml - length(numProd)), numProd];

char_poly = denProd_padded + numProd_padded;

r = roots(char_poly);
disp(r)
```

(d) Now, suppose the plant and controller are given by

$$P(s) = \frac{10}{(s+1)}, \quad C(s) = \frac{K}{s}. \quad (5)$$

Find the values of K for which the feedback loop is stable.

Ans: We are now given a gain K , and we will have to find the interval of K that makes the feedback system stable where $P(s) = \frac{10}{s+1}$ and $C(s) = \frac{K}{s}$

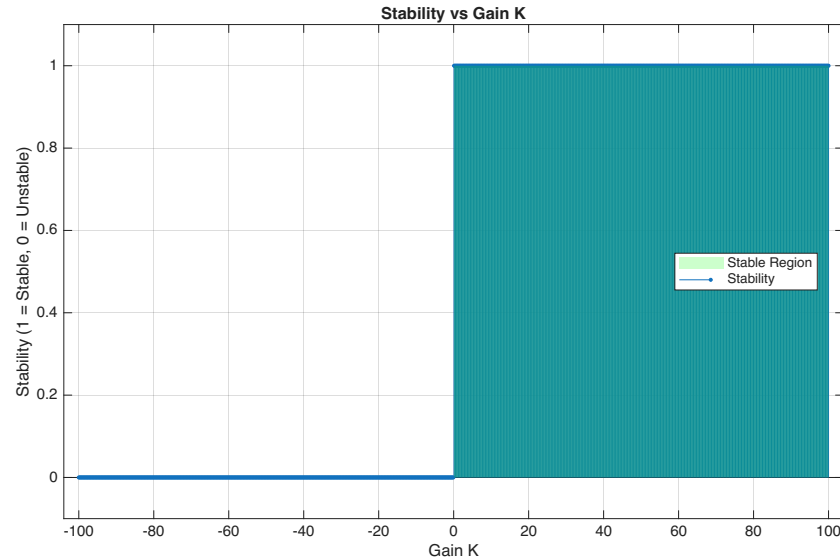
Numerically, I have written a script that defines K as an array of equally spaced points, decided by a discretization step, that increases accuracy the higher it is.

I get from my numeric estimation that for all $0 < K$, the system becomes stable.

The following graph shows a parametrization of this fact, where the gain K is plotted on the x-axis, and stability is true ($= 1$) if K gives stability, and 0 otherwise

Its a step function of sorts

$$y(K) = \begin{cases} 1 & \text{if } K \text{ gives stability} \\ 0 & \text{otherwise} \end{cases}$$



This can also be solved analytically by taking the transfer function, put in the new $P(s)$ and $C(s)$ and solve

$$\begin{aligned}\frac{P(s)}{1 + P(s)C(s)} &= \frac{\frac{10}{s+1}}{1 + \frac{10K}{s^2+s}} \\ &= \frac{(s^2 + s) \frac{10}{s+1}}{s^2 + s + 10K}\end{aligned}$$

In the abc-formula, this becomes

$$\frac{-1 \pm \sqrt{1 - 40K}}{2}$$

If $K = 0$ this gives that $s = 0$ and $s = -1/2$ which gives marginal stability, and for $K < 0$, it gives that s has positive roots. so $K > 0$ gives stability, which is justified by the graph.

The key insight is that K adds phase lag, improving steady state performance

- (e) Fix a value of K such that the system in part (d) is stable, and suppose the system is subject to a disturbance $D(s) = 100$. What will the final value of y be?

Ans: To solve this, we state the final value theorem.

$$\lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} sY(s)$$

we have that

$$Y(s) = \frac{C(s)P(s)}{1 + P(s)C(s)}R(s) + \frac{P(s)}{1 + P(s)C(s)}D(s)$$

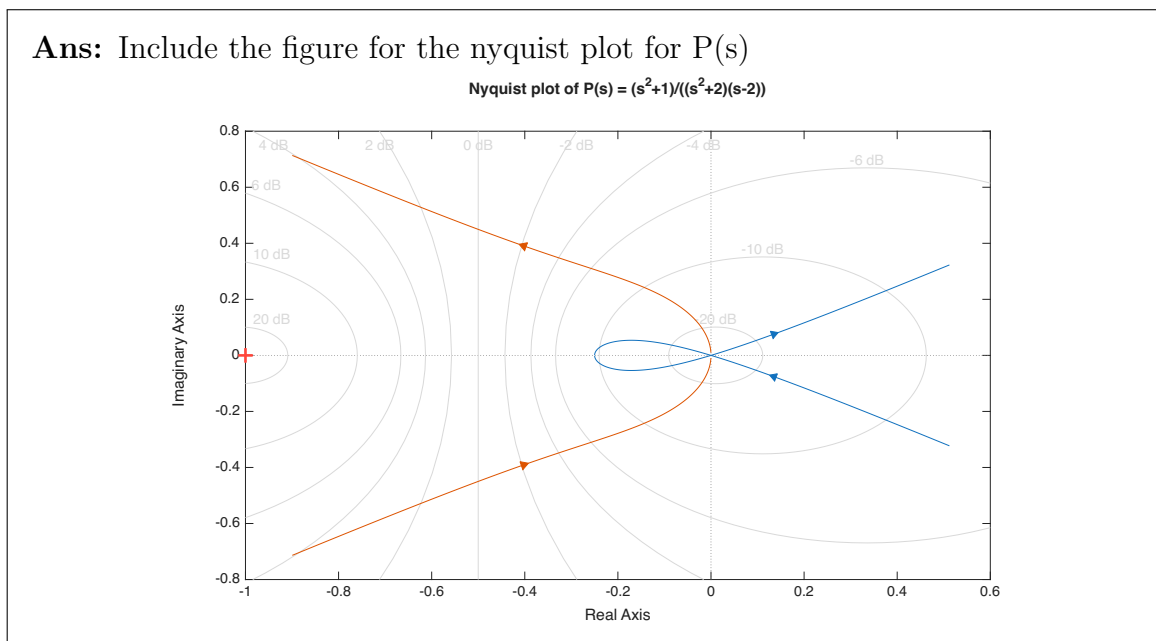
We set $R(s) = 0$ because we don't care about reference when we allow for external disturbances in this case, because we want to examine how disturbances affect our equilibrium state. We also set $D(s) = \frac{100}{s}$ because it is a step function, and $d = 100$ transformed to the Laplace domain, we have that $D(s) = \frac{100}{s}$ so we have

$$\lim_{t \rightarrow \infty} s \frac{\frac{10}{1+s}}{1 + \frac{10K}{s(s+1)}} * \frac{100}{s} = \lim_{t \rightarrow \infty} \frac{1000s}{s^2 + s + 10K} = 0$$

5. (10 points) Consider the standard feedback system with

$$C(s) = K, \quad P(s) = \frac{s^2 + 1}{(s^2 + 2)(s - 2)} \quad (6)$$

(a) Sketch the Nyquist plot of $P(s)$. Include correct asymptotes to infinity and to the origin.



(b) Indicate the number of encirclements (and their orientations) of all possible critical points.

Ans: To indicate the number of encirclements, I want to talk about the Principle of the argument

Suppose we have the D contour. An illustration is given here.

From Classical Control (Francis), its stated that when the characteristic polynomial has no right hand poles (RHP), then the system is stable. It can be shown that for a feedback system like we are dealing with now, the characteristic polynomial is equal to

$$1 + KP(s) = 0$$

Since $C(s) = K$ where K is the gain. The loop gain is defined as the sum of the gain, where gain is explained as the amplified rate of the error between reference and output before its sent into the plant.

- If K is too small, then the system will have large error of steady states and will become too weak
- amplify the gain too much, and the gain is too aggressive, and the loop is destabilized. In our case, the poles of $1 + KP(s) = 0$ moves into the RHP
- At some critical gain, the system is marginally unstable, where the poles are located on the imaginary axis.

Now, the characteristic polynomial is:

$$L(s) = 1 + K * \frac{(s^2 + 1)}{(s^2 + 1)(s - 2)}$$

A fraction = 0 when the numerator = 0

$$\begin{aligned} &= (s^2 + 2)(s - 2) + K(s^2 + 1) \\ L(jw) &= (jw)^3 - (jw)^2(K - 2) + 2(jw) + (K - 4) \\ &= j(2w - w^3) + (-w^2(K - 2) + (K - 4)) \end{aligned}$$

$L(jw)$ is decomposed into a real and imaginary part. We now find the roots of $Im(L(jw))$

$$w(2 - w^2) = 0$$

Which gives that $w = 0$ or $w = \pm\sqrt{2}$. If we then plug $w = 0$ to the real part of $L(jw) = 0$, then we find that $K = 4$. And doing the same for $w = \pm\sqrt{2}$, we find that $K = 0$

In order to indicate the number of encirclements and orientations of all possible critical points, we use the Nyquist stability criterion, which states that

$$Z = N + P$$

where

- Z = number of closed-loop poles in the RHP (i.e. the roots of the characteristic equation)
- N = number of counterclockwise (CCW) encirclements of $\frac{-1}{K}$ by the Nyquist plot.
- P = number of open-loop poles in the RHP (i.e. the poles of $P(s)C(s)$)

We need $Z = 0$ to be true.

We know that there is one open loop pole in the RHP. Namely $s = 2$. So N must equal -1 for stability.

$K = 1$:

From analyzing $P(jw)$ asymptotically, we can find that for $w = 0$, $P(0) = \frac{-1}{4}$. This means that for $K \in (0, 4)$, i.e., $\frac{-1}{K} > \frac{-1}{4}$. As w approaches 1, the numerator of $P(jw)$ goes to 0, and the Nyquist curve crosses the real axis at 0. As w approaches $\sqrt{2}$, the curve goes to ∞ and as w crosses $\sqrt{2}$, the curve goes to $-\infty$. After this, as w goes to ∞ , $P(jw) - \frac{j}{w} = 0$ and it has looped over. What's important to draw here is that for $K \in (0, 4)$, the curve does not cross the critical point $\frac{-1}{K}$ counterclockwise (CCW). And thus, the system is unstable. For $K = 4$, we have reached the marginal stability point, and for $K > 4$, we have stability.

This means that for $K > 4$, we have one CCW encirclement, this $N = -1$, and $Z = -1 + 1 = 0$ and we have stability.

(c) For what range of K is the feedback system stable?

Ans: The feedback system is stable for $K > 4$.

- $K > 4$: Stability
- $K = 4$: marginal stability
- $K \in (0, 4)$: instability

We begin finding stability range by finding what solutions of

$$1 + KP(s) = 0$$

has negative real solutions.

$$1 + KP(s) = 1 + K \frac{(s^2 + 1)}{(s^2 + 2)(s - 2)}$$

$$(s^2 + 2)(s - 2) + K(s^2 + 1) = 0$$

$$s^3 - 2s^2 + 2s - 4 + Ks^2 + K = 0$$

$$s^3 + s^2(K - 2) + 2s + (K - 4) = 0$$

This is pretty hard to solve, but numerically with our matlab solver, we find that the range of stability is

$$[4.0013, 100.0000]$$

When I increase 100 it just becomes the max of the range, so for $K > 4$, we have stability.

This is my output for $K = 4$ in my stability test

```

=== Detailed Analysis for K = 4.0000 ===
Characteristic polynomial: 1.0000 + 2.0000s^2 + 2.0000s
Roots:
  s1 = 0.0000 +0.0000i
  s2 = -1.0000 +1.0000i
  s3 = -1.0000 -1.0000i
SYSTEM IS UNSTABLE

```

If we look at the plot from a), we find that for the critical point $\frac{-1}{K}$, it has one CW encirclement, if $K \neq 4$, and thus $N = -1$ and $Z = N + P = -1 + 1 = 0$ which means the system is stable according to the Nyquist stability criterion.

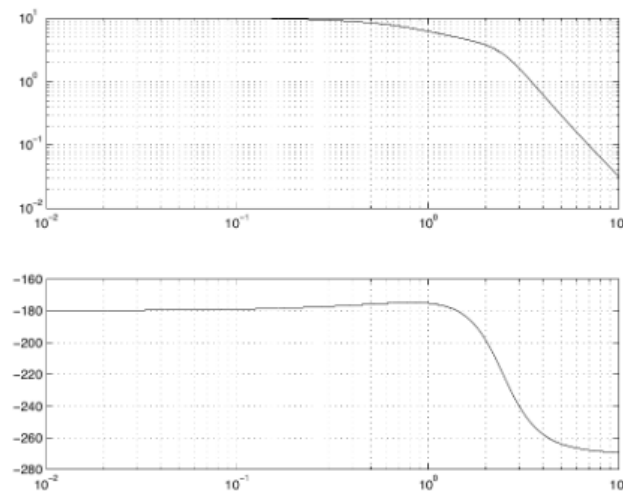


Fig. 1: Bode Plot for Question (6)

6. (15 points) Consider the standard feedback system with $C(s) = K$. The Bode plot of $P(s)$ is given in Figure 1 (magnitude in absolute units, not dB; phase in degrees). The phase starts at -180° and ends at -270° . You are also given that $P(s)$ has exactly one pole in the right half-plane. For what range of gains K is the feedback system stable?

Ans: We have the fundamental equation stemming from the argument of the principle.

$$Z = N + P$$

Where

- Z = Number of closed-loop poles in RHP
- N = number of CCW encirclements of -1 by the Nyquist plot $KP(s)$
- P = number of open-loop poles of $KP(s)$ in RHP

Since we are given that $P = 1$, then $N = -1$ must be true for our feedback system to be stable. This happens when we have a CW encirclement of the critical point. To find this range of K , we can see from the Bode plot that the phase of $P(jw)$, which can be explained as the angle, stays pretty much constant as the frequency approaches 1. When this happens

$$|P(jw_{180})| \approx 1$$

For there to be an encirclement, the point must start to the left of the critical point. i.e. when

$$K\|P(jw)\| = -1$$

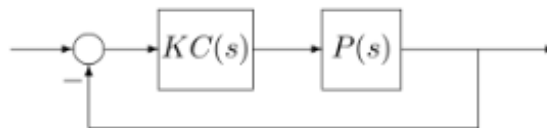
For $K < 1$, we have $|KP(jw)| < 1$ and we will have no encirclements of -1

For $K > 1$, we have that $|KP(jw)| > 1$ and we have an encirclement of -1 , which implies that $N = -1$ and we have

$$Z = (-1) + 1 = 0$$

Which from Nyquist stability criterion entails that the system is stable. The answer to the question of what range of gains K gives stability for the feedback system, the answer is that $K > 1$

7. (10 points) Consider the feedback system:



with

$$P(s) = \frac{s+2}{s^2+2}, \quad C(s) = \frac{1}{s}. \quad (7)$$

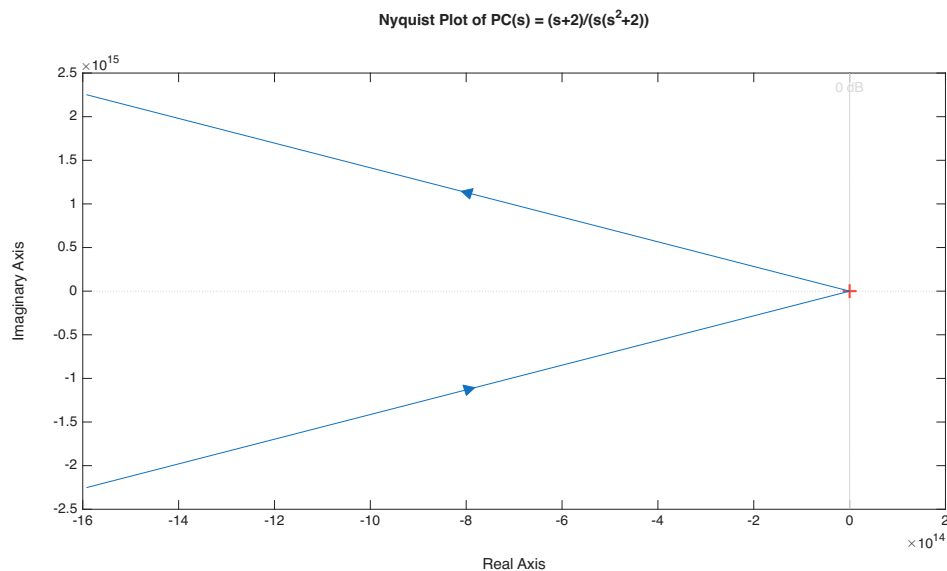
Sketch the Nyquist plot of PC . How many encirclements are required of the critical point for feedback stability? Determine the range of real gains K for stability of the feedback system.

Ans:

$P(s)C(s)$ is the following:

$$P(s)C(s) = \frac{s+2}{s^3+s}$$

Here is the sketch of the Nyquist plot



The denominator of PC is $s(s^2+2)$, and the roots of this equation is

- 0
- $-\sqrt{2}i$
- $+\sqrt{2}i$

Of which, none are in the right hand plane. Therefore, we need no encirclements of the critical point regardless if its counterclockwise or clockwise for the feedback system to be stable.

Furthermore, we can solve for which K that gives stability, by decomposing the equation PC to a real and imaginary part, and solve for which solutions of w that gives no imaginary part.

$$L(s) = KC(s)P(s)$$

$$\begin{aligned} L(s) &= \frac{K}{s} * \frac{s+2}{s^2+2} \\ L(jw) &= \frac{K}{jw} * \frac{jw+2}{2-w^2} \\ &= \frac{K}{2-w^2} * \left(1 - \frac{2j}{w}\right) \\ &= \frac{K}{2-w^2} - j \left(\frac{2K}{w(2-w^2)}\right) \end{aligned}$$

This has no finite solutions for any rational w , only as w approaches ∞ which means that the system is stable for all $K > 0$ $\frac{-1}{K}$ is then in the LHP

8. (10 points) Repeat the previous question with,

$$P(s) = \frac{s^2 + 1}{(s+1)(s^2 + s + 1)}, \quad C(s) = 1. \quad (8)$$

Ans: By plotting the nyquist curve in matlab, we get the Nyquist curve:

$$P(s) = \frac{s^2 + 1}{(s + 1)(s^2 + s + 1)}$$

This plant has poles in

- $s = -1$
- $\frac{-1}{2} + \frac{j\sqrt{3}}{2}$
- $\frac{-1}{2} - \frac{j\sqrt{3}}{2}$

Since all these solutions are in the LHP, then the nyquist plot of $P(s)$ must have no encirclements of the critical point $\frac{-1}{K}$ for the feedback system to be stable. By examining, if we take $K = \frac{-1}{2}$ for instance, we get that the critical point $\frac{-1}{K} = 2$, which the nyquist plot does encircle at all. And for $K > 0$ we can obviously see that this is a stable critical point. Thus, the range

$$(-1, \infty)$$

gives a stable feedback system

Here is the code for calculating the stable K range

```
function stability_analysis_simple()
    P = tf([1 0 1], [1 2 2 1]);
    C = tf([1], [1]);

    K_min = -100;
    K_max = 100;
    n_points = abs(K_max - K_min)*1000; % More points for better numeric approx
    K_values = linspace(K_min, K_max, n_points);

    is_stable = false(size(K_values));
    stability_boundaries = [];

    for i = 1:length(K_values)
        K = K_values(i);

        % Closed-loop system
        T = feedback(K * C * P, 1);
        is_stable(i) = isstable(T);

        % Detect stability boundaries (where stability changes)
```

```

    if i > 1 && is_stable(i) ~= is_stable(i-1)
        stability_boundaries = [stability_boundaries , K_values(i)];
    end
end
stable_indices = find(is_stable);

if isempty(stable_indices)
    fprintf('No stable K-values found in range [%0.2f, %0.2f]\n', K_min, K_max);
else
    K_stable_min = K_values(stable_indices(1));
    K_stable_max = K_values(stable_indices(end));

    fprintf('\n====STABILITY-ANALYSIS-RESULTS====\n');
    fprintf('Stable range: K in [%0.4f, %0.4f]\n', K_stable_min, K_stable_max);
    fprintf('Stability boundaries found at: ');
    if ~isempty(stability_boundaries)
        fprintf('K=%0.4f', stability_boundaries(1));
        for j = 2:length(stability_boundaries)
            fprintf(' , K=%0.4f', stability_boundaries(j));
        end
        fprintf('\n');
    else
        fprintf('None detected in tested range\n');
    end
    fprintf('Number of stable K-values: %d/%d\n', sum(is_stable), length(K_values));
end

% Plot the stability results
figure;
stem(K_values, is_stable, 'filled', 'MarkerSize', 2);
xlabel('Gain K');
ylabel('Stability (1=Stable, 0=Unstable)');
title('Stability vs Gain K');
grid on;
ylim([-0.1 1.1]);

% Highlight stable region
if ~isempty(stable_indices)
    hold on;
    area(K_values(stable_indices), ones(size(stable_indices)), ...
        'FaceColor', 'g', 'FaceAlpha', 0.2, 'EdgeColor', 'none');
    hold off;
    legend('Stability', 'Stable-Region', 'Location', 'best');
end

```

```

% Test and display a few specific points for verification
fprintf( '\n====-VERIFICATION-AT-SPECIFIC-POINTS-====\n' );
test_points = unique([K_min, K_stable_min, 0, K_stable_max, K_max, stability_b]);
test_points = test_points(isfinite(test_points));

for i = 1:min(6, length(test_points)) % Show up to 6 points
    K = test_points(i);
    T = feedback(K * C * P, 1);
    poles = pole(T);

    fprintf( 'K=-%7.3f:- ', K);
    if isstable(T)
        fprintf( 'STABLE--- ');
    else
        fprintf( 'UNSTABLE- ');
    end
    fprintf( 'Poles:- ');
    for j = 1:min(3, length(poles)) % Show up to 3 poles
        fprintf( '%.3f%+.3fi - ', real(poles(j)), imag(poles(j)));
    end
    if length(poles) > 3
        fprintf( '... ');
    end
    fprintf( '\n' );
end
end

stability_analysis_simple()

%%

% Helper function for detailed analysis at specific K values
function analyze_specific_K(num_P, den_P, num_C, den_C, K_values)
    % Precompute the polynomial components
    den_product = conv(den_P, den_C);
    num_product = conv(num_P, num_C);

    % Ensure both polynomials have the same length
    max_length = max(length(den_product), length(num_product));
    den_product_padded = [zeros(1, max_length - length(den_product)), den_product];
    num_product_padded = [zeros(1, max_length - length(num_product)), num_product];

```



```

for K = K_values
    fprintf('\n===== Detailed Analysis for K = %.4f =====\n', K);

    % Form the characteristic polynomial
    char_poly = den_product_padded + K * num_product_padded;

    % Find roots
    r = roots(char_poly);

    % Display results
    fprintf('Characteristic polynomial: ');
    % Create a nice string representation
    poly_str = '';
    for j = 1:length(char_poly)
        if char_poly(j) ~= 0
            if j == 1
                poly_str = sprintf('%.4f', char_poly(j));
            else
                exponent = length(char_poly) - j;
                if exponent == 1
                    poly_str = sprintf('%s + %.4fs', poly_str, char_poly(j));
                else
                    poly_str = sprintf('%s + %.4fs^%d', poly_str, char_poly(j), exponent);
                end
            end
        end
    end
    disp(poly_str);

    fprintf('Roots:\n');
    for j = 1:length(r)
        fprintf('s%d = %.4f + %.4fi\n', j, real(r(j)), imag(r(j)));
    end

    % Check stability
    if all(real(r) < 0)
        fprintf('SYSTEM IS STABLE\n');
    else
        fprintf('SYSTEM IS UNSTABLE\n');
    end
end
end

```

```
% Optional: Analyze specific points of interest  
analyze_specific_K([1, 0, 1], [1, 2, 2, 1], [1], [1, 0], 1);
```